

Q6. EER Model – Generalization, Specialization and Aggregation

Q6

THE THREE LEVELS OF DBMS ARCHITECTURE

DBMS uses a three-level architecture to separate the user's view of data from the physical storage.
This provides data abstraction and data independence.

THE THREE LEVELS

1

EXTERNAL (VIEW) LEVEL

- Also called View Level.
- It represents the user's individual view of the database.
- Different users may have different views.
- It shows only the required part of the database.

2

CONCEPTUAL (LOGICAL) LEVEL

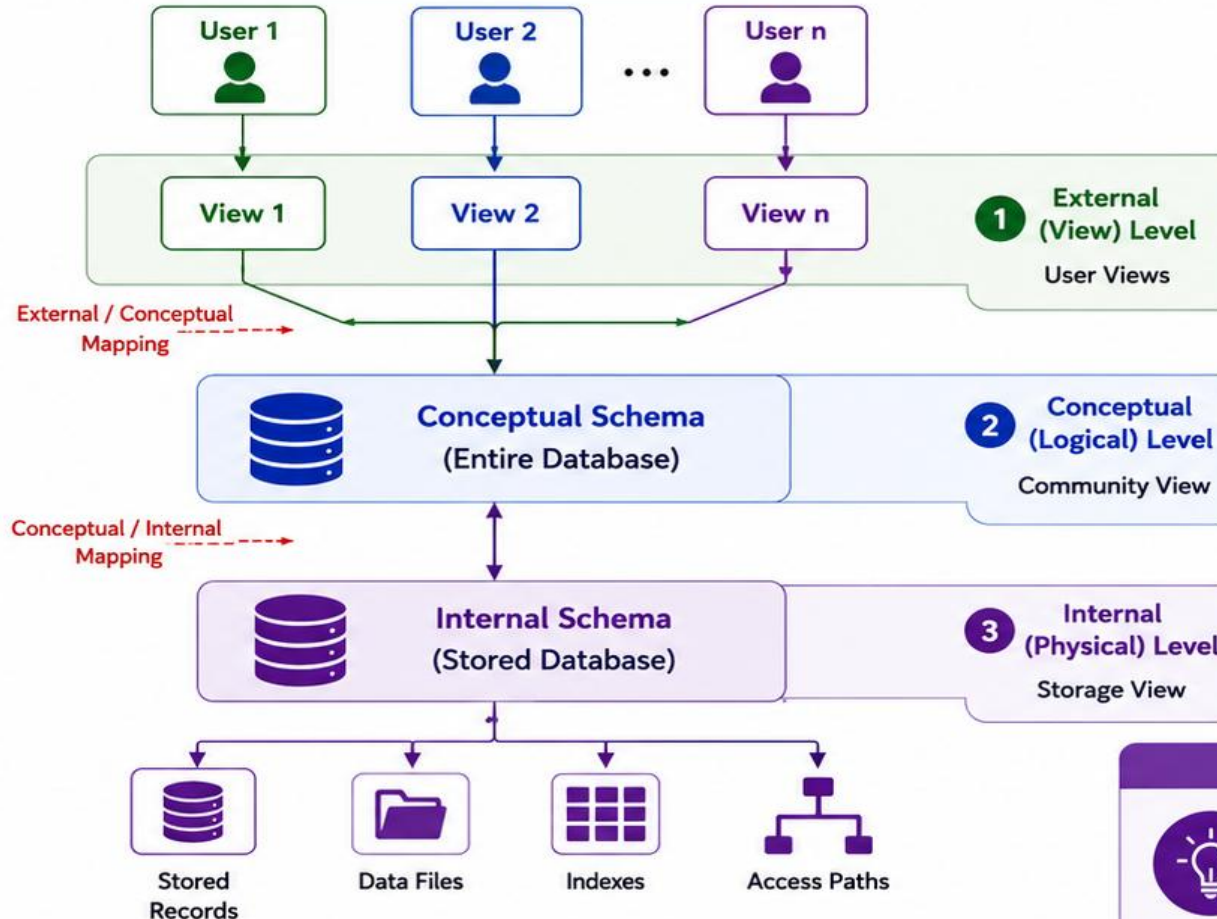
- Also called Conceptual Level.
- It represents the entire database structure.
- Describes what data is stored in the database and the relationships among the data.
- It is independent of physical storage.

3

INTERNAL (PHYSICAL) LEVEL

- Also called Internal Level.
- It describes how the data is actually stored.
- Includes storage structures, access paths and file organization.
- It is independent of the views of users and programs.

ARCHITECTURE DIAGRAM



KEY FEATURES



Data Abstraction
Hides the complexity of the database system from users.



Data Independence
Changes in one level do not affect the other levels.



Multiple User Views
Different users can access the same data in different ways.



Security
Users can be given access only to the data they are authorized to see.



Easier Maintenance
Database changes are easier to manage without affecting user programs.

IN SUMMARY



The three-level architecture provides a clear separation between how users view data, the logical structure of the database, and how data is physically stored.



Key Idea: Three-level architecture → Data Abstraction → Data Independence → Security → Better Performance